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The Editors-in-Chief
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Dear Editors,

We are pleased to submit our manuscript, **"Do topological and quantum-inspired molecular representations add value? Evidence from African antimalarial chemical space"**, for consideration as an Original Research Article in the *Journal of Cheminformatics*.

This work introduces three molecular representations—Topological Fingerprint (TFP), Tensor Network Embedding (TNE), and Quantum Kernel Score (QKS)—and benchmarks them against classical fingerprints on the 19 836 African natural-product-derived antimalarial candidates with complete TNE embeddings; the QKS analysis additionally uses the full 19,849-molecule representation panel. Persistent homology resolves a scaffold paradox (92.6% Tanimoto novelty with 69.3% scaffold recovery) by separating H_0 (connectivity) from H_1 (ring topology). TNE achieves a 6.1× real-atom mean compression while retaining information predictive of computational docking scores on the analysed panel. Quantum kernels, evaluated under a rigorously controlled protocol, showed no statistically detectable difference from a properly tuned classical RBF baseline at the tested sample sizes (6-qubit QKS AUC 0.823 versus RBF 0.829 at $n = 19{,}849$, $p = 0.060$).

The results show that classical ECFP4 remains the strongest single descriptor for primary screening (AUC 0.948). The hybrid descriptor (TFP+TNE+QK) achieves AUC 0.888, significantly below ECFP4 ($p < 0.0001$) yet above every standalone quantum-inspired descriptor, with QKS as the principal positive contributor (ablation $\Delta = -0.040$). A joint analysis with resistance-resilience scores in an expanded computational cohort ($n = 494$) found a weak unadjusted H_1–RRS association ($\rho = 0.2399$) that disappeared after adjustment for molecular size and related structural covariates ($\rho_{\text{partial}} = 0.0329$, $p = 0.4668$), providing no evidence for an independent topological biomarker.

The manuscript addresses a focused methodological question: whether topological and quantum-inspired representations provide predictive gains beyond ECFP4, or instead offer complementary structural diagnosis. The study addresses this question through matched baselines, explicit transfer boundaries, public code, and quantitative reporting of both positive and negative findings. The ChEMBL panel is independent of the eOS80CH label source, but exact molecule-level disjointness from the internal panel was not established; we therefore present it as a label-source-shift transfer analysis rather than an independent molecule-disjoint validation. We believe the work aligns closely with the scope of the *Journal of Cheminformatics*: reusable open-source molecular descriptors, a carefully controlled benchmark, explicit comparability limits, and an assessment showing that QKS did not outperform the tuned RBF baseline under the tested protocols. The quantum kernel did outperform the linear kernel in the primary comparison, so we make no blanket claim about all classical baselines.

Code and machine-readable benchmark results are available in the public repository (`https://github.com/NanaEngo/Malaria_codesV2`) under the MIT licence. This work is original, is not under consideration elsewhere, and all authors have approved the submission.

We thank you for your consideration.

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